Some background Notes on Consumption and Scarring mechanism

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1 Introduction

Understanding how macroeconomic shocks transmit to household behavior is central to the Bank's policy response strategy. Two channels are of particular importance for policymakers: the supply-side energy channel, through which oil price disruptions erode purchasing power and reshape consumption baskets, and the expectations channel, through which shifts in household confidence alter intertemporal spending and saving decisions.

The big questions:

1. Has the response of consumption to the shocks mentioned above changed over time?
2. What happened to consumption following a sharp decline in consumer confidence? (or a rise in inflation expectations)?
3. Is this sufficient to explain consumption weakness in the UK?

Key punchlines to motivate the note (to be refined)

1. Research using US data has shown that the responses of macroeconomic variables to exogenous shocks can change over time.
2. Policymaking therefore needs to look beyond matching policy variables to observed movements in endogenous macroeconomic variables, as turning points play a crucial role.
3. Focusing on turning points can help us understand changes in agents' behaviour that may drive business cycles, including periods of weakness in consumption and GDP.

2 Our empirical framework

This note addresses this question using the heteroskedastic proxy vector autoregression (proxy VAR) framework of Bruns and Lütkepohl (2023). The approach exploits both external instruments and changes in the volatility regime of structural shocks to achieve identification, while simultaneously testing for time variation in the impulse response functions (**That can be key to statistically test that HH consumption has indeed changed after key dates**).

So far, We have run some basic tests using three household-level outcomes, which are the real personal consumption expenditure, the personal saving ratio, and real residential investment. Finally, we want to subject them to two distinct structural shocks: an oil supply news shock and a consumer confidence shock (and the inflation expectations shock from SED colleagues). The latter is not cleaned in the sense it is just the standard deviation of various sub-samples. We will use ita s well and see what the results give.

3 Some thoughts about the shocks / instruments

3.1 Oil Supply News Shock

The first instrument is the oil supply news shock constructed by Känzig (2021). This proxy captures unexpected changes in oil supply expectations, measured as the surprise component of OPEC announcements around scheduled meeting dates. Formally, it is defined as the change in the front-month oil futures price in a narrow window around OPEC announcement dates, ensuring that the measured variation reflects supply-side news rather than demand-side developments. The high-frequency identification strategy provides a compelling instrument: it is plausibly exogenous to contemporaneous macroeconomic conditions and strongly correlated with the underlying structural oil supply shock.

3.2 Consumer Confidence Shock (this one may be quite novel at the Bank - To discuss with SK)

The best way to build the instrument is to retrieve the shock from a small VAR and then use it in the proxy VAR that we have (**already done!**).

3.3 some useful observations

1. There seems to be a good degree of variation in the Gfk headline sample.

2. starting from 1974 gives us room to test a lot different macro/political shifts/regimes.

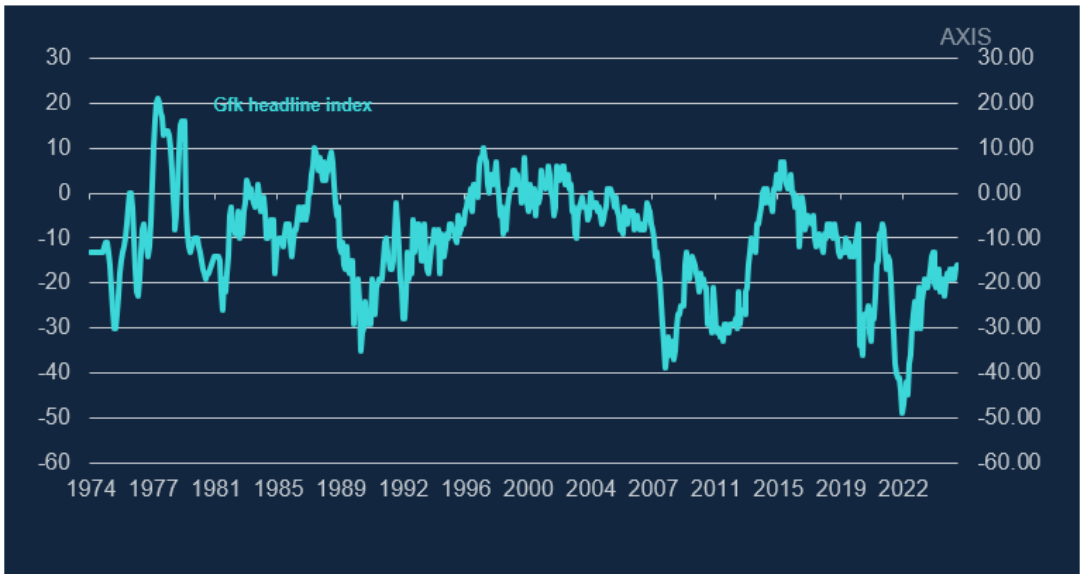

Figure 1: UK Consumer Confidence (GfK Headline Index)

3. I can think of 2008 and 2020 as key dates to pay attention to (To be discussed with SK).

3.4 step 1

We estimate a monthly trivariate VAR in log-levels with 12 lags and seasonal dummies:

$$Y_t = c + \sum_{j=1}^{12} A_j \, Y_{t-j} + D_m + u_t \tag{1}$$

where $Y_t = (c_t, \ ip_t, \ conf_t)'$ contains log real per capita consumption, log industrial production, and the Gfk confidence index, and D_m denotes monthly seasonal dummies. We order $conf_t$ last in the system and apply a Cholesky decomposition to the reduced-form covariance matrix, $\Sigma_u = PP'$, so that the orthogonalized innovation

$$\tilde{u}_t^{E5Y} = \left(P^{-1} u_t\right)_3 \tag{2}$$

captures movements in consumer confidence that are not explained by contemporaneous variation in consumption or industrial production. This orthogonalization ensures that the confidence proxy is purged of the mechanical contemporaneous feedback from real activity, providing a cleaner measure of exogenous shifts in household expectations about future economic conditions.

This shock should be correct. If not we can use the pre-cleaning technique of Agrippino and Rico (2024) as a robustness check (**trick to be tested on Friday after the NTO**).

4 The Heteroskedastic Proxy VAR

In the second stage, each instrument $z_t \in \{z_t^K,\ \tilde{u}_t^{conf}\}$ is used as an external proxy in the heteroskedastic proxy VAR of Bruns and Lütkepohl (2023). The reduced-form model is:

$$Y_t^* = c + \sum_{j=1}^{p} A_j\, Y_{t-j}^* + D_m + u_t^* \tag{3}$$

where Y_t^* contains the endogenous variables of interest: real personal consumption expenditure, the personal saving ratio, and real residential investment. The structural representation links the reduced-form errors to the structural shocks ε_t through:

$$u_t^* = B_0^{-1}\varepsilon_t \tag{4}$$

Identification of the first column of B_0^{-1} — the impact effects of the shock of interest — relies on the proxy relevance condition:

$$E\left(z_t\, \varepsilon_{1,t}\right) \neq 0 \tag{5}$$

and the exogeneity condition:

$$E\left(z_t\, \varepsilon_{j,t}\right) = 0, \quad j = 2, \ldots, K \tag{6}$$

We can keep this part only for the upcoming internal presentation or in case we would like to turn this work into a working paper or a bank underground post.

5 Data

First set of IRFs.

6 Conclusion

[To be completed.]

References

Barsky, R. B. and E. R. Sims (2012). Information, animal spirits, and the meaning of innovations in consumer confidence. *American Economic Review*, 102(4):1343–1377.

Bruns, M. and H. Lütkepohl (2023). Have the effects of shocks to oil price expectations changed? Evidence from heteroskedastic proxy vector autoregressions. *Working Paper*.

Känzig, D. R. (2021). The macroeconomic effects of oil supply news: Evidence from OPEC announcements. *American Economic Review*, 111(4):1092–1125.